

LAPORAN
TUGAS BASIS DATA



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(2502021)

TI 2A

POLITEKNIK PURBAYA TEGAL
TEKNIK INFORMATIKA 2025/2026

IMPLEMENTASI TRIGGER

1. Jalankan Apache dan MySQL di XAMPP

- Buka **XAMPP Control Panel**
- Klik **Start** pada MySQL

2. Masuk ke Database

Lewat CMD:

```
C:\Windows\System32>cd c:/xampp/mysql/bin

c:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE penjualan_sayur;
Database changed
MariaDB [penjualan_sayur]>
```

3. Buat Trigger

Misalnya trigger untuk mengurangi stok setelah transaksi:

```
MariaDB [penjualan_sayur]> DELIMITER $$
MariaDB [penjualan_sayur]>
MariaDB [penjualan_sayur]> CREATE TRIGGER trg_kurangi_stok
-> AFTER INSERT ON detail_transaksi
-> FOR EACH ROW
-> BEGIN
->     UPDATE sayur
->     SET stok = stok - NEW.jumlah
->     WHERE id_sayur = NEW.id_sayur;
-> END$$
Query OK, 0 rows affected (0.035 sec)

MariaDB [penjualan_sayur]>
MariaDB [penjualan_sayur]> DELIMITER ;
```

4. Cek Trigger

```
MariaDB [penjualan_sayur]> SHOW TRIGGERS;
+-----+-----+-----+-----+-----+-----+-----+-----+
| Trigger | Event | Table | Statement | Timing | Created | sql_mode |
| Definer | character_set_client | collation_connection | Database Collation |
+-----+-----+-----+-----+-----+-----+-----+
| trg_kurangi_stok | INSERT | detail_transaksi | BEGIN
  UPDATE sayur
  SET stok = stok - NEW.jumlah
  WHERE id_sayur = NEW.id_sayur;
END | AFTER | 2026-06-24 20:51:10.34 | NO_ZERO_IN_DATE,NO_ZERO_DATE,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_general_ci |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.020 sec)
```

Hasilnya akan menampilkan trigger yang sudah dibuat.

5. Uji Trigger

Lihat stok sebelum transaksi:

```
MariaDB [penjualan_sayur]> SELECT * FROM sayur;
+-----+-----+-----+-----+
| id_sayur | nama_sayur | harga | stok |
+-----+-----+-----+-----+
| 1 | Bayam | 5000 | 45 |
| 2 | Kangkung | 4000 | 40 |
| 3 | Wortel | 6000 | 30 |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

Tambahkan transaksi dan Lihat stok lagi:

```
MariaDB [penjualan_sayur]> INSERT INTO detail_transaksi
-> (id_transaksi,id_sayur,jumlah,subtotal)
-> VALUES
-> (1,1,5,25000);
Query OK, 1 row affected (0.011 sec)

MariaDB [penjualan_sayur]> SELECT * FROM sayur;
+-----+-----+-----+-----+
| id_sayur | nama_sayur | harga | stok |
+-----+-----+-----+-----+
| 1 | Bayam | 5000 | 40 |
| 2 | Kangkung | 4000 | 40 |
| 3 | Wortel | 6000 | 30 |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

Stok Bayam berubah dari **45 menjadi 40** secara otomatis.

6. Hapus Trigger

```
MariaDB [penjualan_sayur]> DROP TRIGGER trg_kurangi_stok;
Query OK, 0 rows affected (0.003 sec)
```